

Tuning cell adhesion on polymeric and nanocomposite surfaces: Role of topography versus superhydrophobicity



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ABSTRACT

Development of surface modification procedures which allow tuning the cell adhesion on the surface of biomaterials and devices is of great importance. In this study, the effects of different topographies and wettabilities on cell adhesion behavior of polymeric surfaces are investigated. To this end, an improved phase separation method was proposed to impart various wettabilities (hydrophobic and superhydrophobic) on polypropylene surfaces. Surface morphologies and compositions were characterized by scanning electron microscopy and X-ray photoelectron spectroscopy, respectively. Cell culture was conducted to evaluate the adhesion of 4T1 mouse mammary tumor cells. It was found that processing conditions such as drying temperature is highly influential in cell adhesion behavior due to the formation of an utterly different surface topography. It was concluded that surface topography plays a more significant role in cell adhesion behavior rather than superhydrophobicity since the nano-scale topography highly inhibited the cell adhesion as compared to the micro-scale topography. Such cell repellent behavior could be very useful in many biomedical devices such as those in drug delivery and blood contacting applications as well as biosensors.

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1. Introduction

By tuning the surface roughness and composition, the wettability of solid surfaces could be controlled. It is now well-known that enhancement of surface roughness and reduction of surface energy can substantially improve the water-repellency behavior [1–3]. Extremely high water repellent behavior has been referred as superhydrophobicity within the last two decades and numerous papers have been published on this issue due to the promising potential applications of superhydrophobic surfaces in various fields [4–9]. Based on the general definition, a water droplet on a superhydrophobic surface keeps almost its spherical shape with a water contact angle above 150° and sliding angle below 10°; therefore, it can easily roll off when the surface is slightly tilted [10]. Inspired by nature, many researchers suggested different routes for fabrication of superhydrophobic surfaces. Among them, solution-based methods such as the one proposed by Erbil et al. [11] have shown promising potential in large-scale production of superhydrophobic surfaces. Since then, phase separation method has been used for transforming many polymers to superhydrophobic

materials [12,13]; however, low adhesion strength and mechanical durability have always been considered as the main concerns. To overcome such issues, nanoparticles were used instead of non-solvent in the course of fabrication process [7]. The mechanical properties were improved but at the price of adding very high amounts of nanoparticles which is not desirable from the commercial viewpoint. For instance, in the research conducted by Qing et al. [14], the weight ratio of nanoparticles to polystyrene was reported 7:3 for achieving superhydrophobic surfaces. Guo et al. [15] have found that the required nanosilica content should be 50 wt.% with respect to the polymer which is a rather high concentration. Recently, our group suggested a novel method including combined use of non-solvent and nanoparticles to fabricate mechanically stable superhydrophobic films and coatings [6]. In this method, a unique hierarchical morphology is formed on the surface top layer comprising micro-spheres which are decorated with nanoparticles.

Due to the absence of any polar groups in its chemical structure, polypropylene (PP) is an intrinsically hydrophobic polymer with a melting temperature about 170 °C exhibiting different surface topographies based on the employed processing conditions [16]. PP has been widely used in many areas such as home appliances, automotive, and package films because of its balanced physical and chemical properties,

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nontoxicity, and low cost. Therefore, the surface wettability is critical for its practical applications.

In the field of biomedicine and tissue engineering, studying the interactions between cells/proteins and polymeric surfaces is of vital importance from both theoretical and practical viewpoints [17,18]. It is well documented that most known mammalian cells exhibit the tendency to be adhered onto a surface in order to conduct normal metabolism, proliferation and differentiation [19]. It is also well-known that cell adhesion is greatly controlled by surface properties such as surface energy, roughness, and chemical composition [20]. Such complex interrelationships are responsible for the reported controversial behaviors. For example, superhydrophobic surfaces were reported to be extremely cell repellent [21,22]; however, in another work, it was demonstrated that cells not only adhere but also proliferate on superhydrophobic surfaces [23]. Many authors debated on the roles of surface topography and wettability on the cell adhesion efficiency. In a recent publication by our group [9], it was demonstrated that nano-scale topography can result in cell-repellent surfaces. However, in this study, a novel method is proposed to fabricate more uniform superhydrophobic PP surfaces and the main aim is to investigate the effects of surface topography through variation of processing conditions (drying temperature) and material formulation on cell adhesion behavior. Different surface topographies and wettabilities were obtained upon variation of processing conditions and formulation in order to further compare the effects of surface topography and wettability on the cell adhesion behavior.

2. Materials and methods

2.1. Materials

Isotactic polypropylene pellets (weight-average molecular weight (M_w) $\sim 12,000 \text{ g} \cdot \text{mol}^{-1}$ and number-average molecular weight (M_n) $\sim 5000 \text{ g} \cdot \text{mol}^{-1}$) were purchased from Sigma-Aldrich with melting point of 157°C and density of 0.9 g/mL . The unmodified silica nanoparticles with specific surface area of $170\text{--}200 \text{ m}^2/\text{g}$ and primary particle size of $15\text{--}20 \text{ nm}$ were procured from US Research Nanomaterials, Inc. Isopropyl alcohol (IA) and p-Xylene were supplied from Merck and used as received.

2.2. Preparation of PP coatings

Solution casting technique was used for fabrication of all the samples in this study. In the case of PP coatings without nanoparticles, 100 mg PP granules were dissolved in 5 mL p-Xylene at elevated temperatures. After achieving homogeneity, several drops of PP solution were casted onto the pre-cleaned glass slides. After 10 min , the whole ensemble was immersed into the non-solvent (isopropyl alcohol) bath for different immersion times. Afterwards, the coatings were dried at two different temperatures in a vacuum oven (25 and 100°C) in order to attain different surface topographies. For fabrication of nanocomposite coatings, different weight percentages of nanosilica were added to the PP solution and then further stirred for 3 h followed by ultrasonic mixing for 30 min in order to achieve a proper level of nanoparticles dispersion within the system. Similar to the pure PP coatings, several drops of nanosilica/PP suspension was casted onto the substrates and then they were immersed into the non-solvent bath for the optimum immersion time of 5 min . For comparison reasons, one nanocomposite sample was also prepared in the absence of non-solvent bath. To this end, after obtaining the suspension of PP and nanosilica, a few drops were casted on the substrate and then left to dry at room temperature in a vacuum oven for 24 h .

For simplicity, the prepared non-solvent-based samples are referred under the general code of IAX in which IA accounts for isopropyl alcohol and X signifies the immersion time. The nanocomposite samples are also named under the general code of IAXSiY in which Y is indicative of the weight percentage of added nanoparticles. Due to the concurrent

use of nanoparticles and non-solvent and their synergistic effect, the required amount of nanoparticles for achieving superhydrophobicity was highly diminished. Based on the preliminary experiments, $10 \text{ wt.}\%$ of silica nanoparticles were chosen for cell adhesion studies due to its optimal results.

2.3. Characterization

A video-based contact angle measurement system (OCA 15, DataPhysics Instruments GmbH, Filderstadt, Germany) was employed to determine the water contact angle values of the samples. The water contact angle measurements of each sample was conducted at least three times across the sample surface using the sessile drop method by dispensing $4 \mu\text{L}$ drops of de-ionized water (surface tension $\gamma_{\text{lv}} = 72.8 \text{ mN} \cdot \text{m}^{-1}$ at 25°C) on the samples surfaces. All water contact angle values were measured under ambient laboratory conditions at a temperature of 25°C . Morphologies of the coating surfaces were evaluated on a digital scanning electron microscope (KYKY-EM3200, KYKY Technology Development, Beijing, China) operated at 25 kV . To avoid electric charging all samples were plated with gold coating. All the roughness parameters were acquired for $260 \times 260 \mu\text{m}^2$ field size surfaces by means of a 3D confocal microscope $\mu\text{surf explorer}$ (NanoFocus AG, Oberhausen, Germany). X-ray photoelectron spectroscopy (XPS) analysis was performed by using XPS spectroscopy with a monochromatic AlK_{α} X-ray source (1486.6 eV photons), operated at 180 W (12 kV and 15 mA) and under ultra-high vacuum conditions.

2.4. Cell-culture experiments

The 4T1 mouse mammary tumor cell line (estrogen receptor (ER)/progesterone receptor (PR)-negative) (National cell bank of Iran, Pasteur institute, Tehran, Iran) was cultured in RPMI 1640 medium (Gibco BRL Inc., Grand Island, NY, USA), supplemented with adding 10% heat-inactivated fetal bovine serum, 1% penicillin/streptomycin, 1 mM pyruvate and 2 mM L-glutamine, and maintained in humidified incubator at 37°C in a $5\% \text{ CO}_2$ atmosphere. Tumor cell suspensions of greater than 90% viability were prepared from subconfluent cultures with 0.25% trypsin (Invitrogen, CA, USA) and 0.02% Ethylenediaminetetraacetic acid (EDTA).

To prepare samples for SEM observation, at the end of incubation (24 h), samples were washed with Phosphate-buffered saline (PBS) to remove unattached cells. The adhered cells were then fixed by glutaraldehyde solution (2.5%) for 10 min and dehydrated through a series of ethanol/water mixtures ($10, 30, 50, 70, 80, 90$, and 100%) each for 10 min . Dehydration was followed by air drying overnight.

2.5. MTT assay

Cell proliferation was quantified by means of MTT assay, a standard colorimetric assay, in which yellow MTT (3-(4,5-Dimethylthiazol-2-yl)-2,5 diphenyltetrazolium bromide, a tetrazole) is reduced to purple formazan. Briefly, the coating samples were washed with phosphate buffer, transferred into new plates containing $5:1$ ratio of media and MTT solution (5 mg/mL in Phosphate buffer), respectively, and incubated for 2 h at 37°C . After removing the culture media, 0.5 mL of dimethyl sulfoxide (DMSO) was added to dissolve the insoluble purple formazan product into a colored solution. The absorbance of the supernatant was recorded by a microplate reader at 540 nm . Cell population was determined through a standard curve which was established through a certain number of cells counted by a Neubauer-counting chamber. This test was done according to the procedure presented by Zund et al. who have developed a linear correlation between the cell number and the MTT absorbency at 550 nm [24].

3. Results and discussion

3.1. Wettability behavior

To evaluate the hydrophobic properties, water contact angles and sliding angles were determined for all the samples and illustrated in Fig. 1. In the case of pure PP coatings, it can be seen that immersing the as-casted PP solution into the non-solvent bath has increased the water contact angles significantly, which could be attributed to the proper level of induced surface roughness onto the coatings' surfaces. According to Fig. 1, although the water contact angle substantially increases upon immersing the PP coating inside the non-solvent bath for 1 min, superhydrophobicity is not yet achieved. It could be inferred that the time needed for solvent/non-solvent exchange and interaction of PP chains with non-solvent molecules was not provided in this case. As the immersion time is increased to 5 min, the water contact angle is highly increased above 150° and the sliding angle is reduced to below 10° , which is in accordance with the general definition of superhydrophobicity [10]. It was found that longer immersion times were not in the favor of water repellency; for example, despite the fact that the water contact angle was obtained above 150° for IA10, its sliding angle was increased; and also, the sample suffered from surface inhomogeneity. Therefore, the optimum immersion time could be chosen as 5 min.

In order to study the effect of drying temperature on the surface wettability, the optimum sample was also dried at an elevated temperature, and it was named IA5_100. It is seen that increasing the solvent evaporation rate resulted in lower water repellency behavior, and IA5_100 cannot be regarded as a superhydrophobic material due to its rather high sliding angle.

With the aim of studying the effects of variation in formulation, nanocomposite samples were also prepared. At first, the effect of nanoparticles addition in the absence of non-solvent bath was studied. To this end, 10 wt.% nanosilica was introduced to the PP solution followed by a thorough dispersion, and then, casted under ambient conditions. It was found that the water contact angle was insignificantly increased from 104° to 116° , and no trace of superhydrophobicity was observed. However, once the non-solvent bath step was included into the above process, a considerable increase was observed for the water contact

angle, and thus, a superhydrophobic surface was obtained; however, the sliding angle value was about 10° . One can infer that combined use of nanoparticles and non-solvent bath acted synergistically in the favor of superhydrophobicity which is in accordance with the previous results [6].

3.2. Surface morphology and topography

In order to find out the reasons for the obtained wettability results, surface morphology of samples was investigated by scanning electron microscopy. Fig. 2 displays the SEM images of samples made by the individual use of non-solvent bath. It is seen that the surface of IA1 sample was smoother and less porous as compared with IA5 and IA10, which can be attributed to the very short immersion time. In fact, the immersion time was too short for PP macromolecules to establish interactions with isopropyl alcohol molecules. Although the induced level of roughness was not enough to achieve superhydrophobicity but it increased the water contact angle significantly as compared to the utterly smooth PP surface (from 104° to 136°).

In contrast, as the immersion time is increased to 5 min, a substantial change can be detected in the morphology of sample. According to Fig. 2c, an interconnected network comprising of ultra-fine microspheres, formed an ordered and uniform surface morphology. Moreover, numerous surface voids were formed on the top layer where air could be enclosed. Upon increasing the immersion time to 10 min, a rather similar morphology to that of IA5 can be seen such that the interconnected network maintains its structure, albeit with much finer spheres which are difficult to be detected (Fig. 2b). This difference could be the reason for the slightly lower level of water-repellency behavior in the case of IA10 as compared to IA5.

An interesting observation was made once the system was dried at an elevated temperature. Considering Fig. 2c and d, a comparison could be made between IA5_room and IA5_100 which were dried at room and elevated temperatures, respectively. It is evident from Fig. 2d that much bigger spheres were formed and some bare areas can be detected leading to an utterly different surface topography for IA5_100 as compared to IA5_room. The reason for the observed phenomenon could be attributed to the much higher evaporation rate due to the enhanced drying temperature. The effect of evaporation rate on the surface morphology of polymeric films has been previously investigated [25]. It is known that the phase separation at the top layer of the coating occurs differently than the bulk due to the air/solid interface at the coating's top layer. As the solvent starts to evaporate, polymer chains, which were previously swollen by solvent molecules, tend to shrink and form micro-spheres due to the fact that the sphere shape has the lowest surface energy and thus the highest level of stability among other geometries. It can be postulated that once the PP macromolecules had enough time in the course of drying (IA5_room), the size of as-formed spheres becomes as small as possible in order to increase the system stability. On the other hand, in the case of IA5_100, due to the very high drying temperature which is even higher than boiling points of the used solvent and non-solvent, the PP chains had a very limited time to deform and shrink as much as possible leading to formation of bigger micro-spheres. This marked difference between surface topographies will be used for studying cell adhesion behavior in the next section.

Fig. 3 displays the surface morphologies of nanocomposite coatings made by different routes. It is reminded that 10 wt.% nanosilica was used for both samples. Based on Fig. 3a, it is clear that the amount of added nanosilica did not cover the whole surface completely. Such orderless distribution of nanoparticles and exposure of smooth PP surface are the reasons for the observed low increase in water contact angle (116°). In contrast, once the PP suspension containing 10 wt.% nanosilica was immersed into the non-solvent bath for 5 min, the nanosilica particles completely covered the top layer of the coating resulting in a uniform distribution according to Fig. 3b. The interesting

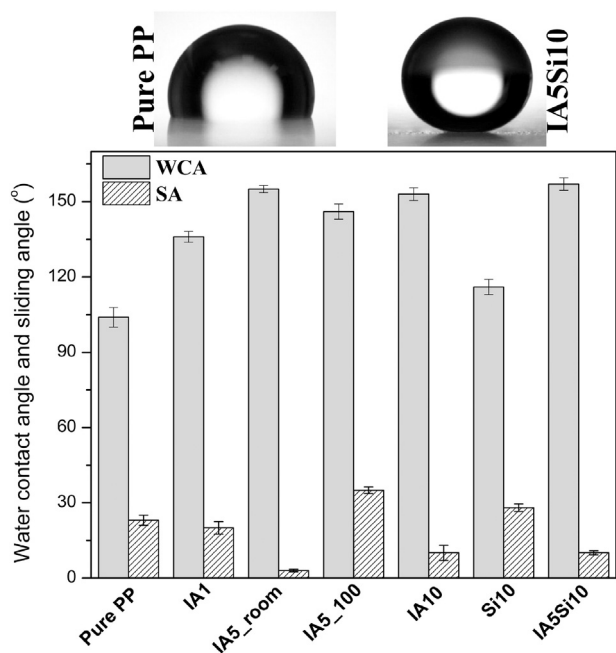


Fig. 1. Water contact angle (WCA) and sliding angle (SA) values for all the samples. Insets are the water drop profiles for two samples.

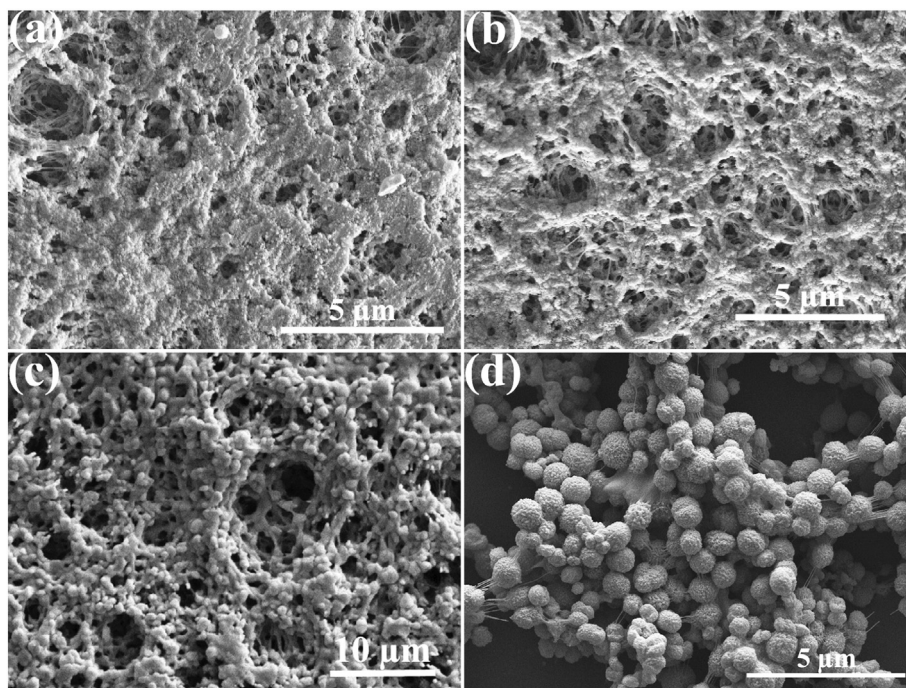


Fig. 2. Surface morphologies obtained by SEM analysis for (a) IA1, (b) IA10, (c) IA5_room, and (d) IA5_100.

point is that the nanoparticles content is the same for both samples. It can be speculated that use of non-solvent bath instantly caused the phase separation to occur at the depth of the coating leading to dividing the system into two phases, namely, poor-PP and rich-PP phases [7]. PP-rich phase forms the dense layer at the depth of the coating, and PP-poor phase, in which nanoparticles are more present, forms the top layer of the coating resulting in a uniform distribution of nanosilica particles on the surface. The induced level of roughness together with self-assembly of nanosilica particles on the top layer are the main reasons for the observed superhydrophobic behavior. Moreover, a different topography is obtained for IA5Si10 which could be used for cell adhesion studies.

The topography of Si10 and IA5Si10 was studied by confocal microscopy, and the obtained results are shown in Fig. 4. As can be seen, the topography images confirm the obtained morphologies from SEM results. The sample made in the absence of non-solvent exhibits a non-uniform surface topography indicating that the surface was not covered completely by nanoparticles. On the other hand, once non-solvent bath was included in the preparation process (IA5Si10), an utterly packed topography was obtained which could be the major reason for the observed superhydrophobic behavior. The quantified results were also attained. The average roughness (R_a) for Si10 and IA5Si10 samples were measured 2.8 and 4.1 μm , respectively. Moreover, the developed interfacial area ratio, namely Sdr, were 1.8 and 2.47 for Si10 and

IA5Si10 samples, respectively. In fact, Sdr is expressed as the percentage of additional surface area contributed by the texture as compared to an ideal plane [7]. These results further confirm that a rougher structure has been formed once the coating was immersed within the non-solvent bath. In fact, this modified process results in aggregation and self-packing of nanoparticles at the top layer of the coating which is the major reason for the observed superhydrophobic behavior.

3.3. Surface composition

Apart from the surface topographical features, surface chemical composition is also highly influential on the superhydrophobicity of samples. Therefore, X-ray photoelectron spectroscopy (XPS) was employed to determine the surface chemical compositions. To this end, Si10 and IA5Si10 samples were evaluated by XPS and the results are shown in Table 1. First observation is that a combination of C, Si and O are simultaneously present at the coatings' top layers. Once the non-solvent bath step was not included in the preparation of nanocomposite coatings (Si10), the majority of the coating's top layer was comprised from PP macromolecules. Oppositely, with including non-solvent bath step in the process (IA5Si10), the Si atomic percentage was highly increased from 8 to 17.5%, and the atomic percent of C significantly decreased implying the strong presence of silica nanoparticles on the surface. It is obvious that due to the molecular structure of

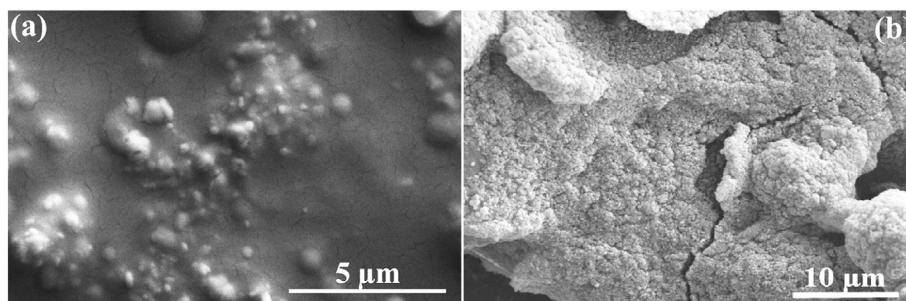


Fig. 3. SEM images of (a) Si10 and (b) IA5Si10.

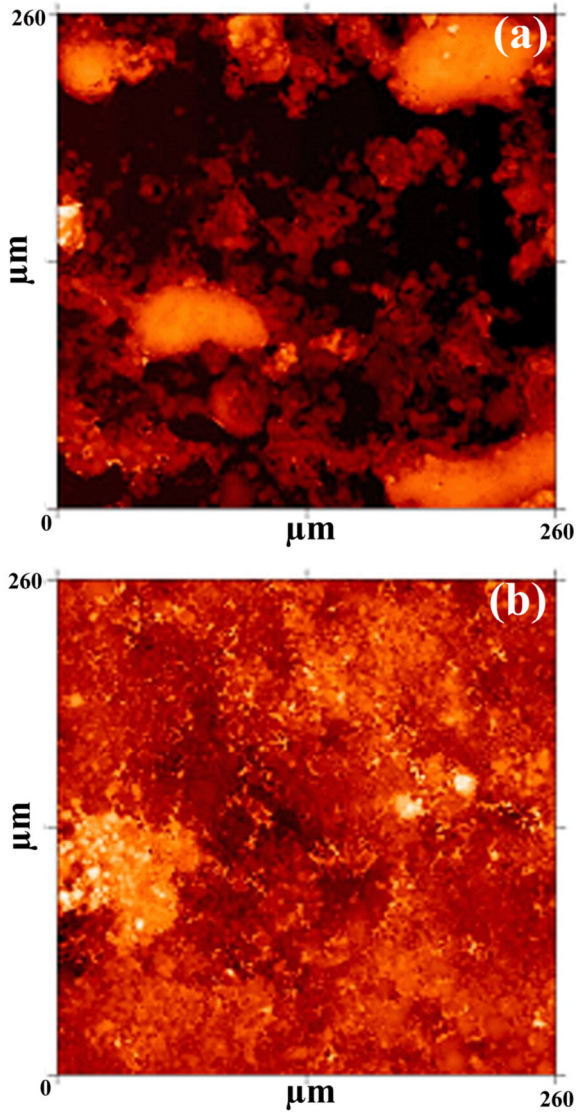


Fig. 4. Topography images for (a) Si10 and (b) IA5Si10.

SiO₂, the [O]/[Si] should be equal to 2; however, this ratio is slightly higher than 2 for both samples. It was earlier mentioned that the used silica in this study is not a modified grade and thus hydroxyl groups might be present on its surface. This claim can be confirmed by the wettability results obtained in the case of IA5Si10 sample for which the sliding angle was about 10 in spite of the very high water contact angle. It could be speculated that those hydroxyl groups on the surface of silica might have established polar interactions with the water droplet leading to reduction of self-cleaning behavior.

3.4. Theoretical considerations

Two models have been generally used for describing different wettability behaviors. The first is Wenzel model [1], according to which, water contact angle on a rough surface made by material with intrinsic water contact angle higher than 90° should be enhanced

towards superhydrophobicity values while a rough surface with an intrinsic water contact angle lower than 90° tends to superhydrophilicity. In this model, the water droplets penetrate into the pores and cannot roll off the surface easily. The Wenzel model of wetting on rough surfaces utilizes the roughness factor (r), as the ratio of the actual surface area to the projected planar area, to describe the correlation between the observed water contact angle value measured on the rough surface (θ_w), and the equilibrium water contact angle of the flat surface (θ_E). Moreover, this model is valid only when there is no gas (air) entrapped beneath the water droplet.

$$\cos \theta_w = r \cos \theta_E. \quad (1)$$

In contrary, according to Cassie–Baxter model [2], since the air enclosed between the coating surface and the water droplet minimizes the contact area, hydrophobicity is increased by enhancing the surface roughness. Under such circumstances, the apparent contact angle (θ_{CB}) is given according to the Cassie–Baxter equation:

$$\cos \theta_{CB} = f_{solid} \cos \theta_{solid} + f_{air} \cos \theta_{air}. \quad (2)$$

By substituting $f_{air} = 1 - f_{solid}$, $\theta_{air} = \pi$, and assuming that the water contact angle of the solid fraction (θ_{solid}) corresponds to the equilibrium water contact angle of the smooth surface (θ_E), Eq. (2) can be rearranged as:

$$\cos \theta_{CB} = f_{solid} (\cos \theta_E + 1) - 1. \quad (3)$$

Eq. (3) can also be rewritten for the solid fraction (f_{solid}):

$$f_{solid} = \frac{1 + \cos \theta_{CB}}{1 + \cos \theta_E}. \quad (4)$$

According to the wettability results in the previous section, IA1 and IA5 samples conform the Wenzel regime due to the fact that the water droplet is partially penetrated into the surface grooves leading to rather high sliding angle values. On the other hand, Cassie–Baxter state was detected for IA5_{room} and IA5Si10 samples due to the very high level of air entrapment within the surface voids and pores. In fact, if a pure Cassie–Baxter model is assumed for IA5_{room}, the value for f_{solid} would be calculated as 0.12 implying that 88% of the water droplet is in contact with air instead of solid surface.

3.5. Cell adhesion study

The main aim of this research was to induce different surface topographies and wettabilities in order to study the cell adhesion behavior. To this end, 4T1 mouse mammary tumor cells were employed. Scanning electron microscopy (SEM) analysis was also utilized since this approach is able to distinguish substrate-dependent differences in the shape of individual cells in high-density cultures. For pure PP coating, which has been prepared by melting PP between two glass slides, a moderate level of cell spreading was detected (SEM not shown here). This level of cell adhesion was reduced upon induction of a micro-sized topography to the coating surface in the case of IA5_{room}. The attached cells shown in Fig. 5a was detected based on a large scanned area. However, once the cell culture period was further extended to 7 days, the extent of cell attachment was highly increased and it can be noticed that a lot of cells were growing on which leads to cover almost the entire surface topography. In fact, after seven days of incubation, most of the cells were extensively spread with flattened nuclear region. The increased cell adhesion was even more profound than that for the pure PP coating. It could be inferred that at the early stages of cell culture period, the extremely low level of surface energy of superhydrophobic sample delayed the cell attachment; however, the induced micro-sized roughness together with the presence of large voids and pores were more dominative than the effect of surface energy

Table 1
XPS atomic contents (at.%) for the nanocomposite coatings.

Samples	[C]	[O]	[Si]	[O]/[Si]
Si10	73	17.5	8	2.18
IA5Si10	32.5	47	20.5	2.3

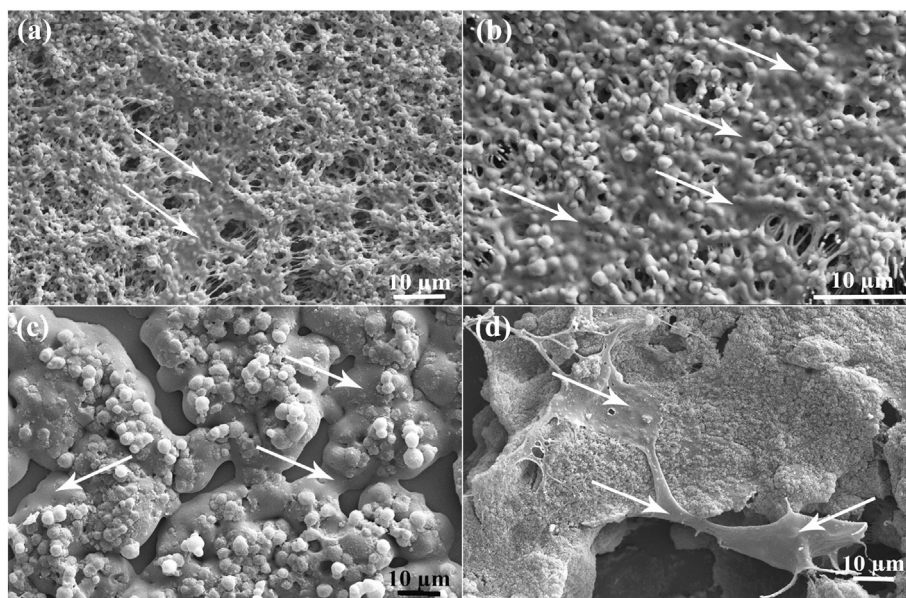


Fig. 5. SEM images for cell cultured (a, b) IA5_room, (c) IA5_100 and (d) IA5Si10. Note that (a), (c) and (d) are the SEM images after 1 day of culture but (b) is SEM image after 7 days.

leading to a much higher contact area between cells and the solid surface.

Comparing Fig. 5a and c, one could see the effect of surface topography on the cell attachment behavior. It is clear that the level of adhered cells on the surface is severely increased for IA5_100 as compared with IA5_room. It was previously shown that increasing the drying temperature, or in other words, enhancement of evaporation rate affected the surface topography and much larger micro-spheres were formed on the coating's surface. Here, the surface energy factor can be neglected due to the small difference in water contact angles (9°). On the other hand, surface topography becomes an influential factor due to the observed difference in their morphologies (Fig. 2). It now can be concluded that surface topography plays a more important role in cell adhesion behavior of PP coatings rather than superhydrophobicity. In fact, the effect of surface topography in controlling the cell adhesion was previously studied by Ventre et al. on different micro-patterned surfaces [26].

The interesting observation was made once the nanosilica particles were also included in the preparation process (IA5Si10). A very large scan area was performed in SEM analysis and no adhered cells were observed on the majority of the coating surface. Despite the obvious reduction in cell spreading, cells could be still observed on the surface of IA5Si10 sample (Fig. 5d). However, cell colonization was sparse and an actin cytoskeleton was developed, albeit rather poorly established which is indicative of bad adherence to the surface. In fact, cells were preferentially attached to the crevices between the surface asperities whereas other areas of the surface were proved to be cell-free. Therefore, one can conclude that nano-scale roughness severely impeded the cell adhesion. Such result was also reported by Yang et al. who found that gallium nitride nanostructures reduced cell adhesion as compared to the flat surfaces. They attributed this phenomenon to the nano-structure and also surface wettability [27].

To quantitatively show the precise levels of cell adhesion on the samples, the cytocompatibility of samples was evaluated via MTT assay at 1 and 7 days after the seeding. According to Fig. 6, pure PP nearly maintained its cytocompatibility during the culture period. It also can be seen that cell adhesion was notably reduced on the surface of IA5_room at the earlier stages of cell culture period; however, the cells were highly adhered and proliferated once the culture period extended to 7 days. As earlier stated, there exist opposing factors governing the cell adhesion behavior for this sample: low surface energy which is dominative at the earlier stages of culture, and micro-sized rough

structure which accelerated cell attachment due to the increase in contact area between the cells and solid surface.

In the case of IA5_100, the level of cell adhesion to the surface is even higher than that of the pure PP surface. The effect of surface topography on cell adhesion can be emphasized here. In fact, larger micro-sized rough structure, formed in this case, is the main reason for the observed cell adhesion behavior. As opposed to other samples, IA5Si10, with a nano-scale rough structure and superhydrophobic behavior, exhibits the highest level of resistance against cell attachment and proliferation even after a long time (7 days of culture). Such result can be most probably attributed to the inability of cells to spread along such water repellent coating. However, the role of ultra-fine nano-scale rough structure should not be overlooked in reducing the contact area between cells and the solid surface.

4. Conclusions

A novel method including concurrent use of nanoparticles and non-solvent bath was employed in this study to investigate the effects of different surface topographies and wettabilities on the cell adhesion behavior. To this end, different wettabilities including hydrophobic

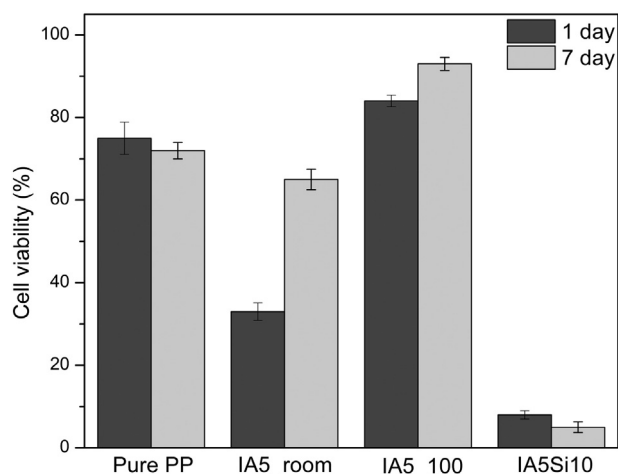


Fig. 6. Cell viability for different samples at 1 and 7 days of postseeding.

and superhydrophobic PP surfaces were obtained through the proposed method. The optimum immersion time was determined and the corresponding as-casted solution was also dried at elevated temperatures. An utterly different surface morphology was attained upon drying at elevated temperatures which was also responsible for the observed loss in water repellency behavior of the coating. It was found that the superhydrophobic sample which was prepared in the absence of nanoparticles (IA5_{room}) exhibited a higher water-repellency behavior rather than the superhydrophobic nanocomposite sample (IA5Si10). However, cell adhesion was much more suppressed in the case of latter sample as compared to the former one, which was attributed to the nano-scale induced roughness in the case of nanocomposite sample. Another comparison was also made for the samples dried at different temperatures. According to the obtained results, it is suggested that surface topography plays a more crucial role in cell adhesion behavior of polymeric surfaces as compared to the surface energy, or in other words, superhydrophobicity.

Acknowledgments

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